University of Ottawa

Department of Mathematics and Statistics

MAT 1302C: Mathematical Methods II

Professor: Hadi Salmasian

Third Midterm Exam – Version A

Novembre 20, 2018

Surname _____ First Name _____

Student # _____ DGD _____

Instructions:

(a) You have 80 minutes to complete this exam.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

(g) Cellular phones, unauthorized electronic devices or course notes (unless an open-book exam) are not allowed during this exam. **Phones and devices must be turned off and put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

By signing below, you acknowledge that you have ensured that you are complying with the above statement.

Signature _____

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	3	4	3	4	6	3	23
Grade							

1. **[3 points]** For each of the following sets, write "**Yes**" if the set is a subspace of $\mathbb{R}^n$ for the given value of n, and write "**No**" if it is not a subspace of $\mathbb{R}^n$. You will receive .5 points for each correct answer, 0 points for each answer left blank, and -.5 points for each incorrect answer. Your total score for this question will not be negative.

_____ A circle in $\mathbb{R}^2$, centred at the origin and with radius equal to 1, $n = 2$.

Solution: No

_____ The set $\left\{ \begin{bmatrix} 3a+2b \\ -3a+b \\ a+b \\ a-b \end{bmatrix} \;\middle|\; a,\, b \in \mathbb{R} \right\}$, $n = 4$.

Solution: Yes

_____ The set Col(A) where A is a 5×2 matrix, $n = 5$.

Solution: Yes

_____ The set $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; x,\, y,\, z \in \mathbb{R} \text{ and } 2x+3y=z \text{ and } y+2z=5x \right\}$, $n = 3$.

Solution: Yes

_____ The set Span$\{v_1, v_2, v_3\}$ for vectors $v_1, v_2, v_3 \in \mathbb{R}^4$, $n = 4$.

Solution: Yes

_____ The set $\left\{ \begin{bmatrix} x \\ y \end{bmatrix} \;\middle|\; x,\, y \in \mathbb{R} \text{ and } x-y=2 \right\}$, $n = 2$.

Solution: No

2. [**4 points**] For each of the following statements, determine if it is "**True**" or "**False**". You will receive 1 point for each correct answer, 0 points for each answer left blank, and -.5 points for each incorrect answer. Your total score for this question will not be negative.

_____ For any 4×4 matrix A, we have $\det(-A) = \det(A)$.

Solution: True

_____ For any $n \times n$ matrix A, if $\det(A) = 0$ then A is invertible.

Solution: False

_____ The equation $x^4 + 3x^3 + 10x + 7 = 0$ does not have any roots in complex numbers.

Solution: False

_____ The rank of an invertible $n \times n$ matrix is always equal to n.

Solution: True

3. **[3 points]** Compute the determinant of the following matrix using the row reduction method.

$$A = \begin{bmatrix} 3 & -5 & 2 & 1 \\ -3 & 5 & -6 & 7 \\ 0 & -1 & 4 & -9 \\ 3 & -3 & 2 & 5 \end{bmatrix}$$

Solution:

$$\begin{aligned}
\begin{vmatrix} 3 & -5 & 2 & 1 \\ -3 & 5 & -6 & 7 \\ 0 & -1 & 4 & -9 \\ 3 & -3 & 2 & 5 \end{vmatrix} &= \begin{vmatrix} 3 & -5 & 2 & 1 \\ 0 & 0 & -4 & 8 \\ 0 & -1 & 4 & -9 \\ 3 & -3 & 2 & 5 \end{vmatrix} && (L_2 \to L_2 + L_1) \\
&= \begin{vmatrix} 3 & -5 & 2 & 1 \\ 0 & 0 & -4 & 8 \\ 0 & -1 & 4 & -9 \\ 0 & 2 & 0 & 4 \end{vmatrix} && (L_4 \to L_4 - L_1) \\
&= -\begin{vmatrix} 3 & -5 & 2 & 1 \\ 0 & -1 & 4 & -9 \\ 0 & 0 & -4 & 8 \\ 0 & 2 & 0 & 4 \end{vmatrix} && (L_2 \leftrightarrow L_3) \\
&= -\begin{vmatrix} 3 & -5 & 2 & 1 \\ 0 & -1 & 4 & -9 \\ 0 & 0 & -4 & 8 \\ 0 & 0 & 8 & -14 \end{vmatrix} && (L_4 \to L_4 + 2L_2) \\
&= -\begin{vmatrix} 3 & -5 & 2 & 1 \\ 0 & -1 & 4 & -9 \\ 0 & 0 & -4 & 8 \\ 0 & 0 & 0 & 2 \end{vmatrix} && (L_4 \to L_4 + 2L_3) \\
&= -3 \cdot (-1) \cdot (-4) \cdot 2 \\
&= -24
\end{aligned}$$

4. Let A, B and C be 2×2 matrices such that

$$\det A = \frac{1}{3} \quad \text{and} \quad \det B = 2 \quad \text{and} \quad 5A^{\mathrm{T}}B^2C^{\mathrm{T}}A^{-1} = I,$$

where I denotes the 2×2 identity matrix.

a) **[3 points]** Calculate $\det C$.

Solution: We have

$$\det(5A^{\mathrm{T}}B^2C^{\mathrm{T}}A^{-1}) = \det(I) = 1.$$

On the other hand, for the left hand side of the above we have

$$\begin{aligned}\det(5A^{\mathrm{T}}B^2C^{\mathrm{T}}A^{-1}) &= 5^2 \det(A^{\mathrm{T}}) \det(B^2) \det(C^{\mathrm{T}}) \det(A^{-1}) \\ &= 25 \det(A) \det(B)^2 \det(C) \det(A)^{-1} \\ &= 25 \det(B)^2 \det(C) \\ &= 25 \cdot 4 \det(C) \\ &= 100 \det(C).\end{aligned}$$

It follows that $100 \det(C) = 1$ so that $\det(C) = \frac{1}{100}$.

b) **[1 point]** Is C invertible? Justify your answer.

Solution: Yes, C is invertible because $\det(C) \neq 0$.

5. Consider the matrix $A = \begin{bmatrix} 1 & -2 & 1 & 2 & 4 & -1 \\ -2 & 4 & -5 & 5 & -2 & -4 \\ 2 & -4 & 2 & 4 & 8 & -2 \end{bmatrix}$.

a) **[4 points]** Find a basis of $\mathrm{Nul}(A)$.
b) **[1 point]** Find a basis of $\mathrm{Col}(A)$.
c) **[1 point]** Calculate the rank of A.

Solution:

a)

$$\begin{bmatrix} 1 & -2 & 1 & 2 & 4 & -1 \\ -2 & 4 & -5 & 5 & -2 & -4 \\ 2 & -4 & 2 & 4 & 8 & -2 \end{bmatrix} \xrightarrow{L_2 \to L_2 + 2L_1} \begin{bmatrix} 1 & -2 & 1 & 2 & 4 & -1 \\ 0 & 0 & -3 & 9 & 6 & -6 \\ 2 & -4 & 2 & 4 & 8 & -2 \end{bmatrix} \xrightarrow{L_3 \to L_3 - 2L_1}$$

$$\begin{bmatrix} 1 & -2 & 1 & 2 & 4 & -1 \\ 0 & 0 & -3 & 9 & 6 & -6 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{L_2 \to -\frac{1}{3}L_2} \begin{bmatrix} 1 & -2 & 1 & 2 & 4 & -1 \\ 0 & 0 & 1 & -3 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{L_1 \to L_1 - L_2}$$

$$\begin{bmatrix} 1 & -2 & 0 & 5 & 6 & -3 \\ 0 & 0 & 1 & -3 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Therefore the general solution of the linear system $Ax = 0$ is given by

$$\begin{cases} x_1 = 2x_2 - 5x_4 - 6x_5 + 3x_6 \\ x_3 = 3x_4 + 2x_5 - 2x_6 \end{cases} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -5 \\ 0 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -6 \\ 0 \\ 2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_6 \begin{bmatrix} 3 \\ 0 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Thus the vectors $\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}; \begin{bmatrix} -5 \\ 0 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}; \begin{bmatrix} -6 \\ 0 \\ 2 \\ 0 \\ 1 \\ 0 \end{bmatrix}; \begin{bmatrix} 3 \\ 0 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ form a basis of $\mathrm{Nul}(A)$.

b) From the REF of A that was computed above, it follows that the 1st and the 3rd columns of A are the pivot columns. Therefore we obtain a basis of $\mathrm{Col}(A)$ consisting of the vectors $\left\{ \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}; \begin{bmatrix} 1 \\ -5 \\ 2 \end{bmatrix} \right\}$.

c) The rank of A is equal to the dimension of $\mathrm{Col}(A)$, so that $\mathrm{rank}(A) = 2$.

6. [**3 points**] Calculate the determinant of the following matrix.

$$A = \begin{bmatrix} \frac{1}{i+2} & 1+3i \\ i & 5-5i \end{bmatrix}$$

Solution:

$$\begin{aligned} \begin{vmatrix} \frac{1}{i+2} & 1+3i \\ i & 5-5i \end{vmatrix} &= \frac{5-5i}{i+2} - (1+3i)i \\ &= \frac{(5-5i)(2-i)}{(2+i)(2-i)} - (i-3) \\ &= \frac{10-5i-10i-5}{4-i^2} - i + 3 \\ &= \frac{5-15i}{5} - i + 3 \\ &= 1 - 3i - i + 3 \\ &= 4 - 4i \end{aligned}$$